

Near-Square-Root NP-Hardness for Euclidean CVP Would Collapse the Exponential Hierarchy

A Barrier Theorem and a Conditional Near-Matching Fixed-Exponent Compiler

somewhereafter

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Abstract

Aggarwal, Bennett, Brakerski, Golovnev, Kumar, Li, Peters, Stephens-Davidowitz, and Vaikuntanathan proved a quantitative public-coin protocol placing rank- D Euclidean GapCVP $_\gamma$ in coAM with complexity $2^{O(D/\gamma^2)}$ up to polynomial factors. We derive two consequences for hardness near the square-root scale. First, if, for any fixed $C > 0$, GapCVP $_{\sqrt{D}/\log^C D}$ is NP-hard under deterministic polynomial-time promise-preserving many-one reductions, then the exponential hierarchy collapses to its second level. More generally, the same conclusion holds for a reduction from length- N inputs whenever $D(N)/\gamma(D(N))^2 = \log^{O(1)} N$. Second, under AMETH, a fixed-exponent reduction producing rank $D(N) = N^{K+o(1)}$ and hardness factor D^c must satisfy $K(1 - 2c) \geq 1$ in exponent order.

We then give a constructive counterpart. Conditional on an explicit rectangular matrix-valued Frobenius–Hankel common-atom lemma, we modify the folded-moment reduction of Mira by an elementary multiplication-friendly alphabet reduction. Large-field axis labels are represented by Reed–Solomon style evaluations over a subfield; one fixed linear decoder recovers every bounded-degree product needed by the compiler. Crucially, the same decoder preserves the pointwise “bits pay for rank” factorization for arbitrary tables, so no inner-code consistency theorem is required. For every fixed $c < 1/2$ and $\varepsilon > 0$, the resulting conditional reduction has

$$D(N) \leq N^{(5+\varepsilon)/(1-2c)+o(1)}.$$

Thus its output-degree dependence improves from $\Theta((1 - 2c)^{-2})$ to $\Theta((1 - 2c)^{-1})$, matching the AMETH lower bound up to a constant arbitrarily close to five. We do not claim a uniform $D^{1/2-o(1)}$ hardness theorem, hardness at $\sqrt{D}/\text{polylog } D$, or a machine-checked proof of the new algebraic lemma.

Claim boundary. The hierarchy-collapse and AMETH barrier results in Part I are unconditional implications from published theorems. The alphabet-reduced compiler in Part II is conditional on [Theorem 7.1](#). The multiplication-friendly encoding, rank factorization, and parameter ledger are proved here; the common-atom statement currently has a research proof architecture but has not been independently verified or formalized in Lean. No dimension-dependent near-square-root NP-hardness result is claimed.

Contents

1	Introduction	2
2	Preliminaries	4
2.1	Promise reductions	4
2.2	Quasipolynomial and exponential resources	4
2.3	AMETH	5

3	The quantitative coAM protocol	5
4	The hierarchy-collapse barrier	5
4.1	The exact frontier of this argument	6
4.2	Why the protocol does not yield a deterministic algorithm	7
5	The stronger AMETH boundary and the protocol deficit	7
6	Where the existing folded-moment reduction loses efficiency	8
7	Algebraic input: a rectangular common-atom lemma	8
8	Multiplication-friendly alphabet reduction	10
8.1	Evaluation–interpolation encoder	10
8.2	The alphabet-reduced table	10
8.3	Completeness	11
8.4	High-rank soundness	12
8.5	Low-rank soundness	12
9	The parameter ledger	12
10	A conditional near-matching fixed-exponent theorem	13
11	Why the decoder does not need to authenticate inner codewords	15
12	Architectural optimality and routes below the constant five	15
13	The uniform near-square-root frontier	16
14	Uniformity and verification boundary	16
14.1	Deterministic field construction	16
14.2	What is and is not checked	16
15	Conclusion	17

1 Introduction

The Euclidean closest vector problem asks for the distance from a target $t \in \mathbb{Q}^D$ to a full-rank lattice $L \subseteq \mathbb{R}^D$. Its promise version GapCVP_γ distinguishes

$$\text{dist}(t, L) \leq r \quad \text{from} \quad \text{dist}(t, L) > \gamma(D)r.$$

The square-root scale is a persistent complexity boundary. Goldreich and Goldwasser gave a polynomial-time coAM protocol around $\sqrt{D/\log D}$ approximation, while Aharonov and Regev placed $\sqrt{D}$ -approximate CVP in $\text{NP} \cap \text{coNP}$. Mira recently proved, in a Lean-formalized deterministic reduction, NP-hardness at D^c for every fixed $c < 1/2$ [11]. The qualifier “fixed” is essential: the polynomial degree of the emitted rank is allowed to depend on c and diverges as $c \uparrow 1/2$.

A natural next target is a single dimension-dependent factor such as

$$\gamma_c(D) = \frac{\sqrt{D}}{\log^c D}. \tag{1.1}$$

The first part of this paper records an architecture-independent obstruction to polynomial-time NP-hardness at (1.1). Aggarwal et al. prove the quantitative protocol

$$\text{GapCVP}_\gamma \in \text{coAM-TIME}\left(2^{O(D/\gamma^2)} \text{poly}(M)\right), \quad (1.2)$$

where M is the target encoding length [1]. Substitution of (1.1) makes the protocol quasipolynomial. Composing it with a hypothetical NP-hardness reduction and applying the resource-scaled Boppana–Håstad–Zachos collapse theorem of Pavan, Selman, Sengupta, and Vinodchandran collapses the exponential hierarchy to level two [12, 5].

The same quantitative protocol gives a more refined efficiency constraint. For a reduction from source size N , define the *protocol deficit*

$$\Xi(N) := \frac{D(N)}{\gamma(D(N))^2}. \quad (1.3)$$

The protocol complexity is $2^{O(\Xi(N))} \text{poly}(M(N))$. Under AMETH, a polynomial-output reduction cannot have $\Xi(N) = o(N)$ on sparse 3-CNF instances. In particular, if

$$D(N) = N^{K+o(1)}, \quad \gamma(D) = D^c,$$

then AMETH forces

$$K(1 - 2c) \geq 1 \quad (1.4)$$

in exponent order. This converts the qualitative statement “the output degree must diverge” into a quantitative lower bound on how fast it must diverge.

The second part asks whether Mira’s fixed-exponent reduction can approach this lower bound. The original folded-moment compiler stores a large-field label $u \in \mathbb{F}_Q$ explicitly at every sample point. This one-hot dictionary costs Q^r labels, even though a satisfying witness activates only one. The key new idea is to separate the field used for polynomial semantics from the field used for one-hot labels. Write

$$F_0 = \mathbb{F}_q, \quad F = \mathbb{F}_{q^\nu},$$

with fixed extension degree ν . An element of F is encoded by evaluations of its degree- $< \nu$ representative polynomial at ℓ points of F_0 . Coordinatewise products of up to d encoded elements are decoded by one fixed linear map $F_0^\ell \rightarrow F$. In an honest table this recovers exactly the large-field folded products. For an arbitrary table, the fixed decoder passes through the moment sums and yields the same rank factorization used in the high-rank soundness proof. Therefore invalid inner encodings need not be recognized: algebraically rich decoded tables are expensive, and algebraically poor decoded tables enter the low-rank atomization argument.

Conditional on a matrix-valued common-atom lemma, this removes the N^r label-dictionary loss. The resulting family has

$$K(1 - 2c) \leq 5 + \varepsilon, \quad (1.5)$$

for arbitrary fixed $\varepsilon > 0$. Combined with (1.4), this gives an upper/lower efficiency theory within a constant factor:

$$1 \leq K(1 - 2c) \leq 5 + \varepsilon. \quad (1.6)$$

The dependence on $1 - 2c$ is optimal under AMETH, although the constant five is not claimed to be universally optimal.

Contributions. The paper has five principal contributions.

1. We prove the source-sensitive hierarchy-collapse criterion $D(N)/\gamma(D(N))^2 = \log^{O(1)} N$ and its fixed-polylogarithmic-loss consequence.
2. We isolate the exact dimension-only region not covered by that argument and distinguish it from the stronger AMETH barrier.
3. We formulate the protocol deficit (1.3) and derive the fixed-exponent efficiency lower bound (1.4).
4. We give a self-contained multiplication-friendly alphabet reduction and prove that it preserves both completeness and high-rank soundness for the folded-moment compiler.
5. Conditional on the common-atom lemma, we prove the near-matching output-degree theorem (1.5) and show that the constant tends to five, which is optimal inside a natural full-matrix moment architecture.

Organization. Sections 2–5 contain the barrier results. Sections 6–10 develop the conditional compiler. Section 12 explains the constant five and the remaining routes for improvement. Section 13 returns to the dimension-dependent frontier, and Section 14 records the verification boundary.

2 Preliminaries

2.1 Promise reductions

A rank- D Euclidean GapCVP instance consists of a full-rank rational lattice basis, a rational target, and a radius $r > 0$. The YES and NO predicates are

$$\text{dist}(t, L) \leq r, \quad \text{dist}(t, L) > \gamma(D)r,$$

respectively. We use deterministic polynomial-time, promise-preserving many-one reductions. If a reduction is applied to source strings of length N , let $M(N)$ denote the bit length of the emitted lattice instance and $D(N)$ its rank. Polynomial time gives

$$D(N) \leq M(N) \leq N^{O(1)}. \tag{2.1}$$

No polynomial lower bound on $D(N)$ is assumed unless explicitly stated.

2.2 Quasipolynomial and exponential resources

Write

$$\text{qpoly}(N) = 2^{(\log N)^{O(1)}}.$$

The class AM_{qpoly} consists of constant-round public-coin protocols whose verifier time, randomness, communication, and prover-message lengths are quasipolynomial. We use the exponential hierarchy

$$\text{EH} = \bigcup_{k \geq 1} \Sigma_k^{\text{EXP}}.$$

The following resource-scaled consequence of the Boppana–Håstad–Zachos argument is recorded in Pavan et al. [12].

Theorem 2.1 (Resource-scaled interactive-proof collapse). *If $\text{coNP} \subseteq \text{AM}_{\text{qpoly}}$, then the exponential hierarchy collapses to its second level. More precisely,*

$$\text{EH} = \text{AMEXP} \subseteq \Sigma_2^{\text{EXP}} \cap \Pi_2^{\text{EXP}}.$$

The proof first scales the Boppana–Håstad–Zachos collapse to quasipolynomial resources and then pads to exponential input lengths; see [12]. We use only this consequence.

2.3 AMETH

Aggarwal et al. formulate the Arthur–Merlin Exponential Time Hypothesis (AMETH) as the assertion that, for some constant $\eta > 0$, sparse 3-TAUT has no constant-round public-coin protocol of complexity $2^{\eta n}$, where n is the number of variables. Together with sparsification, it rules out $2^{o(n)}$ constant-round protocols for sparse 3-TAUT [1, 10]. We use it only as a conditional efficiency benchmark.

3 The quantitative coAM protocol

Aggarwal et al. prove the following time–approximation tradeoff.

Theorem 3.1 (Aggarwal et al.). *For every function $\gamma = \gamma(D) \geq 1$, rank- D Euclidean GapCVP $_\gamma$ has a constant-round coAM protocol of complexity*

$$2^{O(D/\gamma(D)^2)} \text{poly}(M), \quad (3.1)$$

where M is the target-instance encoding length.

The underlying private-coin protocol is a time-scaled version of the Goldreich–Goldwasser translated-ball protocol. The public-coin formulation follows by standard transformations with polynomial overhead; the exponent D/γ^2 is preserved [1, 6, 8, 3].

Writing

$$\gamma(D) = \frac{\sqrt{D}}{f(D)} \quad (3.2)$$

turns (3.1) into

$$2^{O(f(D)^2)} \text{poly}(M). \quad (3.3)$$

This identity is the only lattice-specific input needed for the hierarchy barrier.

4 The hierarchy-collapse barrier

We first state the source-sensitive form.

Theorem 4.1 (General reduction barrier). *Suppose there is a deterministic polynomial-time promise-preserving many-one reduction from SAT instances of length N to Euclidean GapCVP $_\gamma$ instances of rank $D(N)$ and encoding length $M(N)$. If, for some fixed a ,*

$$\frac{D(N)}{\gamma(D(N))^2} \leq (\log N)^a \quad (4.1)$$

for all sufficiently large N , then the exponential hierarchy collapses to its second level.

Proof. By Theorem 3.1, the target promise problem has a coAM protocol of complexity

$$2^{O(D(N)/\gamma(D(N))^2)} \text{poly}(M(N)).$$

Equations (2.1) and (4.1) make this quasipolynomial in N . Composing the verifier with the deterministic reduction gives

$$\text{SAT} \in \text{coAM}_{\text{qpoly}}.$$

Complementing gives $\text{TAUT} \in \text{AM}_{\text{qpoly}}$, and coNP-completeness of TAUT implies $\text{coNP} \subseteq \text{AM}_{\text{qpoly}}$. Apply [Theorem 2.1](#). $\square$

Corollary 4.2 (Fixed-polylogarithmic loss). *For every fixed $C > 0$, if*

$$\text{GapCVP}_{\sqrt{D}/\log^C D}$$

is NP-hard under deterministic polynomial-time promise-preserving many-one reductions, then the exponential hierarchy collapses to its second level.

Proof. Here

$$\frac{D}{\gamma(D)^2} = \log^{2C} D.$$

Since $D(N) \leq N^{O(1)}$, the right-hand side is $\log^{O(1)} N$. $\square$

Corollary 4.3 (Conditional impossibility). *Assume the exponential hierarchy does not collapse to its second level. Then, for every fixed $C > 0$, Euclidean $\text{GapCVP}_{\sqrt{D}/\log^C D}$ is not NP-hard under deterministic polynomial-time promise-preserving many-one reductions.*

This statement is independent of the internal architecture of a reduction. It applies to codes, PCPs, Fourier statistics, group actions, moment tables, geometric gadgets, and any other polynomial-time compiler.

4.1 The exact frontier of this argument

Under the notation (3.2), [Theorem 4.1](#) applies whenever

$$f(D(N))^2 = \log^{O(1)} N. \quad (4.2)$$

Every dimension-only loss $f(D) = \log^{O(1)} D$ satisfies (4.2) for every polynomial-time reduction. The converse is not asserted: a superpolylogarithmic loss may still be quasipolynomial in the source length if the emitted rank is unusually small.

Under the natural rank-regularity hypothesis

$$D(N) \geq N^\delta \quad (4.3)$$

for some fixed $\delta > 0$, the first dimension-only region not covered by the quasipolynomial-protocol argument is

$$\log^C D = o(f(D)) \quad \text{for every fixed } C.$$

The genuinely near-square-root part of this region additionally has $f(D) = D^{o(1)}$, giving the schematic strip

$$\gamma(D) = \frac{\sqrt{D}}{\log^{\omega(1)} D}. \quad (4.4)$$

There is no smallest superpolylogarithmic function. Examples include $\exp((\log \log D)^{1+\varepsilon})$ and $\exp(\sqrt{\log D})$. Failure of the present argument in this strip is not evidence of hardness; it says only that the published protocol is no longer quasipolynomial after composition.

4.2 Why the protocol does not yield a deterministic algorithm

The conclusion above is an interactive-proof collapse, not a quasipolynomial-time algorithm for GapCVP. The private protocol underlying (3.1) repeats a hidden-bit translated-ball experiment many times. At the near-square-root scale, the number of trials is itself quasipolynomial, so enumerating the private randomness costs exponential time in a quasipolynomial quantity. The Goldwasser–Sipser and Babai–Moran transformations use polynomially more resources than the private protocol but do not produce a logarithmic public seed [8, 3]. Nor may one simply substitute a standard pseudorandom generator: Merlin’s response depends on the revealed public coins, and soundness quantifies over all prover strategies. Thus Theorem 4.2 gives $\text{coNP} \subseteq \text{AM}_{\text{qpoly}}$ under the hypothetical reduction, not $\text{coNP} \subseteq \text{qpoly}$.

5 The stronger AMETH boundary and the protocol deficit

The quantity controlling the protocol is worth naming.

Definition 5.1 (Protocol deficit). For a reduction producing rank $D(N)$ and approximation factor $\gamma(D(N))$, define

$$\Xi(N) := \frac{D(N)}{\gamma(D(N))^2}.$$

Theorem 3.1 gives a protocol of complexity $2^{O(\Xi(N))} \text{poly}(M(N))$. The hierarchy-collapse theorem corresponds to $\Xi(N) = \log^{O(1)} N$; AMETH reaches much farther.

Proposition 5.2 (AMETH barrier). *Assume AMETH. No deterministic polynomial-time reduction from sparse n -variable 3-CNF formulas to Euclidean GapCVP can satisfy*

$$\frac{D(n)}{\gamma(D(n))^2} = o(n) \tag{5.1}$$

on all sparse inputs.

Proof. Equation (5.1) and Theorem 3.1 give a $2^{o(n)}$ constant-round protocol for sparse 3-TAUT. The sparsification lemma lifts this to protocols of complexity $2^{\eta n + o(n)}$ for general 3-TAUT for every fixed $\eta > 0$, contradicting AMETH for sufficiently small η . $\square$

Corollary 5.3 (Fixed-exponent efficiency ceiling). *Assume AMETH. Suppose a fixed-exponent reduction has*

$$D(n) = n^{K+o(1)}, \quad \gamma(D) = D^c, \quad 0 < c < 1/2.$$

If $K(1 - 2c) < 1$, then the reduction contradicts AMETH. Equivalently, any such family must satisfy

$$K \geq \frac{1}{1 - 2c} \tag{5.2}$$

in exponent order.

Proof. The protocol deficit is

$$D^{1-2c} = n^{K(1-2c)+o(1)}.$$

It is $o(n)$ if $K(1 - 2c) < 1$, contradicting Theorem 5.2. $\square$

Writing $c = 1/2 - \delta$, the lower bound is

$$K \geq \frac{1}{2\delta}. \tag{5.3}$$

This does not preclude hardness for every fixed $c < 1/2$: the output-degree parameter may diverge as $c \uparrow 1/2$. It does preclude a polynomial-output uniform $D^{1/2-o(1)}$ theorem under AMETH.

Table 1: Protocol-deficit regimes for polynomial-time reductions. The first two rows are conditional barriers, not unconditional lower bounds.

Deficit $\Xi(N)$	Composed protocol	Consequence
$\log^{O(1)} N$	quasipolynomial AM	EH collapses to level two
$N^{o(1)}$	$2^{N^{o(1)}}$ AM	contradicts AMETH
$N^{\Theta(1)}$	$2^{N^{\Theta(1)}}$ AM	compatible with these barriers

6 Where the existing folded-moment reduction loses efficiency

Mira’s reduction first produces a binary affine system and then applies an exact parity-kernel lattice adapter. The affine compiler has a source size N , a padded Boolean width $n = \Theta(N)$, a radix s , an axis count r , a rank threshold $\kappa = s^r$, and a sample field of size q . In the original ledger one chooses

$$s = N^{a/r+o(1)}, \quad \kappa = N^{a+o(1)}, \quad q = N^{2+a/r+o(1)}.$$

The binary table has one $(n+1) \times (n+1)$ block for every sample and every label in $\mathbb{F}_q^r$. Hence

$$D = N^{2r+4+a(1+1/r)+o(1)}. \quad (6.1)$$

Completeness has weight $R = q(n+1)^2$, while high-rank soundness gives $S = \kappa(q - D_{\deg})$, so

$$\frac{S}{R} = N^{a-2+o(1)}. \quad (6.2)$$

The Euclidean exponent is therefore

$$c_{r,a} = \frac{a-2}{2(2r+4+a(1+1/r))}.$$

The protocol deficit measured in the source exponent is

$$\frac{DR}{S} = N^{2r+6+a/r+o(1)}. \quad (6.3)$$

Near $c = 1/2 - \delta$, optimizing the existing ledger requires $r = \Theta(1/\delta)$ and $a = \Theta(1/\delta^2)$, giving output degree $\Theta(1/\delta^2)$.

The low-rank proof in the original paper atomizes each scalar matrix entry separately. Identifier equations may then involve the union of $n+1$ atom sets, forcing a moment horizon of order $n\kappa$ and a degree bound of order n^2s . A matrix-valued common-atom theorem shortens the horizon to $O(\kappa)$ and the degree bound to $O(nrs)$, improving constants and reducing the output degree by roughly a factor two. It does not by itself remove the N^r factor in the label dictionary, so the asymptotic dependence remains $\Theta(1/\delta^2)$.

The construction below attacks that dictionary factor directly.

7 Algebraic input: a rectangular common-atom lemma

The constructive result uses the following matrix-valued strengthening of the scalar Frobenius–Hankel theorem.

Assumption 7.1 (Rectangular common-atom lemma). Let K be an algebraically closed field of characteristic two and let $M_0, \dots, M_T \in K^{m \times m}$. For integers $k, L \geq 1$, form the block Hankel matrix

$$\mathcal{H}_{k,L} = (M_{a+b})_{(i,a),(j,b)}, \quad 0 \leq a < k, \quad 0 \leq b \leq L.$$

Suppose $\text{rank } \mathcal{H}_{k,L} = t < k$, the entrywise Frobenius identities

$$M_{2e} = M_e^{\circ 2} \quad (0 \leq e < k) \quad (7.1)$$

hold, and the rectangular window contains a protected prefix through Q with at least $2t$ guard columns on both sides; one sufficient numerical condition is

$$L \geq \max\{Q, 2t - 2\} + 2t. \quad (7.2)$$

Then there is a set $U \subset K$ with $|U| \leq t$ and nonzero coefficient matrices $C_\tau \in \mathbb{F}_2^{m \times m}$ such that

$$M_e = \sum_{\tau \in U} C_\tau \tau^e \quad (0 \leq e \leq Q). \quad (7.3)$$

Proof architecture. The block rank gives a t -dimensional minimal partial realization of the matrix sequence, by a rectangular Ho–Kalman stabilization argument [9]. Thus, on the protected prefix,

$$M_e = OA^e R$$

for a t -dimensional state space, with joint reachability and observability. Let $\mathcal{A} = K[A]$. For each matrix entry define the functional

$$\lambda_{ij}(f(A)) = e_i^\top O f(A) R e_j.$$

Minimality makes the family of multiplication pairings $\lambda_{ij}(ab)$ jointly nondegenerate. Since every element of $\mathcal{A}$ has a representative of degree below t , (7.1) implies

$$\lambda_{ij}(a^2) = \lambda_{ij}(a)^2 \quad (a \in \mathcal{A}). \quad (7.4)$$

If $a \in \mathcal{A}$ is nilpotent, then ab is nilpotent for every b . Iterating (7.4) gives $\lambda_{ij}(ab) = 0$ for every i, j, b , and joint nondegeneracy forces $a = 0$. Hence $\mathcal{A}$ is reduced. A finite-dimensional commutative reduced algebra over an algebraically closed field splits as K^u , where $u = \dim \mathcal{A} \leq t$. The coordinates of A are distinct atoms $\tau \in U$. If e_τ is the corresponding primitive idempotent, set

$$C_\tau(i, j) = \lambda_{ij}(e_\tau).$$

Equation (7.4) and $e_\tau^2 = e_\tau$ imply $C_\tau(i, j)^2 = C_\tau(i, j)$, so every entry lies in $\mathbb{F}_2$. Joint nondegeneracy excludes an all-zero coefficient matrix, and spectral reconstruction gives (7.3).

The finite-window realization step is the technical part: the extra guard columns ensure that the shift is well defined on the stabilized state space and that the protected prefix is represented by the same realization. This is the portion that still requires independent verification. The remainder of Part II is conditional on [Theorem 7.1](#).

Compiler consequence. With rank threshold κ , a safe schedule is

$$H_{\text{id}} = \kappa - 1, \quad L = 4\kappa - 4, \quad T = 5\kappa - 5. \quad (7.5)$$

The largest folded digit sum then has total degree at most

$$d = (2r + 3)s, \quad (7.6)$$

and the outer polynomial degree is bounded by

$$D_{\text{deg}} \leq n(2r + 3)s. \quad (7.7)$$

8 Multiplication-friendly alphabet reduction

8.1 Evaluation–interpolation encoder

Let

$$F_0 = \mathbb{F}_q, \quad F = \mathbb{F}_{q^\nu},$$

where the extension degree $\nu \geq 2$ is fixed. Choose an irreducible polynomial $g(Z) \in F_0[Z]$ of degree ν and identify $F = F_0[Z]/(g)$. Every $u \in F$ has a unique representative $f_u(Z) \in F_0[Z]$ of degree below ν .

Fix a multiplication degree d and set

$$\ell = d(\nu - 1) + 1. \quad (8.1)$$

Assume $q \geq \ell$, and choose distinct points $\beta_1, \dots, \beta_\ell \in F_0$. Define

$$\text{Enc} : F \longrightarrow F_0^\ell, \quad \text{Enc}(u)_h = f_u(\beta_h). \quad (8.2)$$

Define a fixed linear map

$$\text{Dec} : F_0^\ell \longrightarrow F \quad (8.3)$$

by interpolating the unique degree- $< \ell$ polynomial through the supplied values and reducing it modulo g .

Lemma 8.1 (Multiplication-friendly decoding). *For any $u_1, \dots, u_k \in F$ with $k \leq d$,*

$$\text{Dec}(\text{Enc}(u_1) * \dots * \text{Enc}(u_k)) = u_1 \cdots u_k, \quad (8.4)$$

where $*$ denotes coordinatewise multiplication. Repeated factors are allowed, so the same identity holds for every monomial of total degree at most d .

Proof. The h th coordinate is the evaluation at β_h of

$$P(Z) = \prod_{j=1}^k f_{u_j}(Z).$$

Its degree is at most $k(\nu - 1) \leq d(\nu - 1) = \ell - 1$. Interpolation therefore recovers P exactly, and reduction modulo g gives the product in F . $\square$

This elementary construction is a multiplication-friendly embedding in the sense used in coding and finite-field multiplication. More elaborate multiplication-friendly and Chudnovsky-type constructions appear in [7, 4, 13]; the simple Reed–Solomon version suffices here because ν is fixed.

8.2 The alphabet-reduced table

Let E/F be the degree- n identifier extension used by the folded-moment compiler. A satisfying assignment determines r axis polynomials

$$Q_{x,j}(X) \in F[X], \quad \deg Q_{x,j} < n, \quad Q_{x,j}(\theta) = t_x^{s_j}.$$

At an outer sample point $p \in F$, the honest axis label is

$$u(p) = (Q_{x,0}(p), \dots, Q_{x,r-1}(p)) \in F^r.$$

Instead of allocating a table block for every label in F^r , encode each axis value by (8.2). For every inner position h , define

$$v_h(p) = (\text{Enc}(Q_{x,0}(p))_h, \dots, \text{Enc}(Q_{x,r-1}(p))_h) \in F_0^r.$$

The new binary variables are blocks

$$A^{p,h,v} = (A_{ij}^{p,h,v})_{0 \leq i,j \leq n} \in \mathbb{F}_2^{(n+1) \times (n+1)},$$

indexed by

$$p \in F, \quad 1 \leq h \leq \ell, \quad v \in F_0^r.$$

Thus the small-field dictionary has q^r labels at each inner coordinate, not q^{vr} labels.

For a multi-index $\alpha \in \mathbb{N}^r$, define the inner point moment

$$\psi_{ij,\alpha}(p, h) = \sum_{v \in F_0^r} A_{ij}^{p,h,v} v^\alpha \in F_0, \quad (8.5)$$

where $v^\alpha = \prod_{j < r} v_j^{\alpha_j}$. Decode across h :

$$\varphi_{ij,\alpha}(p) = \text{Dec}((\psi_{ij,\alpha}(p, h))_{h=1}^\ell) \in F. \quad (8.6)$$

Interpolate these values over $p \in F$ to obtain $\Phi_{ij,\alpha}(X) \in F[X]$, and define the canonical extension-field moments

$$\mu_{ij,e} = \Phi_{ij,\text{dig}(e)}(\theta) \in E. \quad (8.7)$$

Every displayed map is $\mathbb{F}_2$ -linear in the table bits: label powers are fixed coefficients, Dec is F_0 -linear, interpolation is F -linear, quotient evaluation is linear, and squaring in characteristic two is $\mathbb{F}_2$ -linear. Therefore the seven families of folded-moment checks remain ordinary binary affine equations.

8.3 Completeness

Let x satisfy the sparse quadratic source and put $z = (1, x)^\top$. For every (p, h) , activate exactly one small label:

$$A^{p,h,v_h(p)} = zz^\top, \quad A^{p,h,v} = 0 \quad (v \neq v_h(p)). \quad (8.8)$$

Then

$$\psi_{ij,\alpha}(p, h) = z_i z_j \prod_{a < r} \text{Enc}(Q_{x,a}(p))_h^{\alpha_a}.$$

If $|\alpha|_1 \leq d$, [Theorem 8.1](#) gives

$$\varphi_{ij,\alpha}(p) = z_i z_j \prod_{a < r} Q_{x,a}(p)^{\alpha_a}. \quad (8.9)$$

The remainder of the original completeness proof is unchanged: outer interpolation recovers the corresponding product polynomial, evaluation at θ gives $z_i z_j t_x^e$, and the degree, fold, Frobenius, identifier, source, symmetry, and anchor checks hold.

The anchor is imposed at every pair (p, h) :

$$\sum_{v \in F_0^r} A_{00}^{p,h,v} = 1. \quad (8.10)$$

The resulting zeroth inner-moment vector is the all-ones vector $\text{Enc}(1)$, so its decoded value is 1.

There are $q^v \ell$ outer-inner locations, and one active matrix at each. Thus the completeness weight is

$$R = q^v \ell (n+1)^2. \quad (8.11)$$

8.4 High-rank soundness

Write the fixed decoder as

$$\text{Dec}(y_1, \dots, y_\ell) = \sum_{h=1}^{\ell} \lambda_h y_h, \quad \lambda_h \in F. \quad (8.12)$$

The coefficients λ_h do not depend on the queried monomial. This is the key soundness property.

At a fixed outer sample p , define

$$V(v)_a = v^{\text{dig}(a)} \quad (0 \leq a < \kappa), \quad W(v)_b = v^{\text{dig}(b)} \quad (0 \leq b \leq L).$$

The evaluated polynomial block matrix factors as

$$P(p) = \sum_{h=1}^{\ell} \sum_{v \in F_0^r} \lambda_h (I \otimes V(v)) A^{p,h,v} (I \otimes W(v))^T. \quad (8.13)$$

Multiplication by fixed matrices or a scalar does not increase rank, and rank is subadditive. Therefore

$$\text{rank } P(p) \leq \sum_{h,v} \text{rank } A^{p,h,v} \leq \sum_{h,v} \text{wt}(A^{p,h,v}) =: W_p. \quad (8.14)$$

This is exactly the original “bits pay for rank” inequality.

If $P(X)$ has generic rank at least κ , choose a nonzero $\kappa \times \kappa$ minor. Every entry has degree at most D_{deg} , so the determinant-divisor argument bounds the total specialization rank loss by κD_{deg} . Summing (8.14) over the q^ν outer sample points gives

$$\text{wt}(A) \geq \kappa(q^\nu - D_{\text{deg}}). \quad (8.15)$$

Set

$$S := \kappa(q^\nu - D_{\text{deg}}).$$

No consistency constraint is imposed on the inner evaluation vectors. Arbitrary inner tables are sound because only their decoded moments enter the rank dichotomy.

8.5 Low-rank soundness

If the generic rank is below κ , quotient folding produces the same matrix-valued Hankel sequence of canonical moments as in the original compiler. The inner alphabet-reduction layer disappears: only the decoded moments remain. Under [Theorem 7.1](#), the protected prefix has one common atom population of size below κ . The identifier, symmetry, and source equations then force every nonzero coefficient matrix to equal $z_\tau z_\tau^\top$ for a Boolean source solution. The odd anchor ensures that at least one such coefficient matrix exists. Hence an unsatisfiable source has no low-rank affine solution, and every solution has weight at least S .

9 The parameter ledger

Let the encoded source size be N and $n = N^{1+o(1)}$. Fix the field-extension degree $\nu \geq 2$ and the axis count r . Set

$$b := \frac{1}{\nu - 1}, \quad s = N^{b+o(1)}, \quad \kappa = s^r = N^{r/(\nu-1)+o(1)}. \quad (9.1)$$

The inner multiplication degree from (7.6) is $d = (2r + 3)s$, and

$$\ell = (\nu - 1)d + 1 = N^{1/(\nu-1)+o(1)}. \quad (9.2)$$

Choose the small field size q as a power of two with $q = \Theta_{r,v}(s)$ and $q \geq \ell$. The outer sample field has size

$$|F| = q^\nu = N^{\nu/(\nu-1)+o(1)}. \quad (9.3)$$

By choosing the hidden constants in s and q sufficiently large, we may ensure

$$q^\nu \geq 256D_{\deg}$$

with $D_{\deg} \leq n(2r+3)s$.

The table has $(n+1)^2$ matrix entries, q^ν outer samples, ℓ inner coordinates, and q^r small labels. Therefore

$$D = (n+1)^2 q^\nu \ell q^r = N^{K_{r,v}+o(1)}, \quad K_{r,v} = \frac{r+3\nu-1}{\nu-1}. \quad (9.4)$$

Completeness satisfies

$$R = q^\nu \ell (n+1)^2 = N^{(3\nu-1)/(\nu-1)+o(1)}. \quad (9.5)$$

The field slack makes $q^\nu - D_{\deg} = \Theta(q^\nu)$, so

$$S = \kappa(q^\nu - D_{\deg}) = N^{(r+\nu)/(\nu-1)+o(1)}. \quad (9.6)$$

Thus the affine Hamming gap exponent relative to D is

$$d_{r,v} = \frac{r-2\nu+1}{r+3\nu-1}, \quad (9.7)$$

and the Euclidean exponent after the parity-kernel adapter is

$$c_{r,v} = \frac{r-2\nu+1}{2(r+3\nu-1)}. \quad (9.8)$$

The protocol deficit has source exponent

$$\frac{DR}{S} = N^{\Delta_v+o(1)}, \quad \Delta_v = \frac{5\nu-2}{\nu-1} = 5 + \frac{3}{\nu-1}. \quad (9.9)$$

Most importantly, Δ_v is independent of r .

The exact identity

$$K_{r,v}(1-2c_{r,v}) = \Delta_v = 5 + \frac{3}{\nu-1} \quad (9.10)$$

connects the construction directly to the AMETH lower bound.

10 A conditional near-matching fixed-exponent theorem

Theorem 10.1 (Conditional efficient fixed-exponent hardness). *Assume [Theorem 7.1](#). For every fixed c with $0 < c < 1/2$ and every fixed $\varepsilon > 0$, there is a deterministic polynomial-time, promise-preserving many-one reduction from 3-SAT of encoded size N to integer-target Euclidean GapCVP $_{D^c}$ such that*

$$D \leq N^{(5+\varepsilon)/(1-2c)+o(1)}. \quad (10.1)$$

The output lattice is the same parity-kernel lattice as in Mira's reduction; in particular, after coordinate permutation its basis entries lie in $\{0, 1, 2\}$.

Table 2: Output-degree behavior near $c = 1/2 - \delta$. The middle two rows use the rectangular common-atom lemma; the final row is the AMETH barrier.

Construction or barrier	Output-degree scale
Original folded-moment compiler	approximately $2/\delta^2$
Common-atom horizon only	approximately $1/\delta^2$
Alphabet reduction with $\nu = 2$	approximately $4/\delta$
Alphabet reduction, optimized fixed ν	approximately $2.5/\delta$
AMETH lower bound	$0.5/\delta$

Proof. Choose a fixed extension degree ν large enough that

$$\frac{3}{\nu-1} < \frac{\varepsilon}{2} \quad \text{and} \quad \frac{1}{\nu-1} < \frac{\varepsilon}{2(1-2c)}. \quad (10.2)$$

For this ν , choose the least integer r with $c_{r,\nu} > c$. Since

$$1 - 2c_{r,\nu} = \frac{\Delta_\nu}{K_{r,\nu}},$$

minimality of r and the fact that increasing r increases $K_{r,\nu}$ by $1/(\nu-1)$ give

$$K_{r,\nu} \leq \frac{\Delta_\nu}{1-2c} + \frac{1}{\nu-1} \leq \frac{5+\varepsilon}{1-2c}. \quad (10.3)$$

The alphabet-reduced compiler has completeness threshold R and soundness threshold S from (9.5)–(9.6). Equation (9.8) and the strict inequality $c_{r,\nu} > c$ give $S > D^{2c}R$ after fixed constant slack and power-of-two rounding. The all-integer parity-kernel identity then converts affine Hamming weight into squared Euclidean distance, proving D^c hardness. For fixed r, ν , all field sizes and table dimensions are polynomial in N . $\square$

Corollary 10.2 (Efficiency sandwich under AMETH). *Assume AMETH and Theorem 7.1. Let $K^*(c)$ denote the infimum of constants K for which a deterministic reduction with output rank $D(N) \leq N^{K+o(1)}$ proves D^c -hardness. For every fixed $c < 1/2$, in exponent order,*

$$\frac{1}{1-2c} \leq K^*(c) \leq \frac{5+\varepsilon}{1-2c}$$

for every fixed $\varepsilon > 0$.

Writing $c = 1/2 - \delta$, Theorem 10.1 gives

$$D \leq N^{(5+\varepsilon)/(2\delta)+o(1)}, \quad (10.4)$$

while AMETH requires $D \geq N^{1/(2\delta)-o(1)}$ in exponent order.

Numerical example. For $c = 1/4$, the original displayed construction uses output exponent 47. The common-atom horizon alone gives approximately 35. The alphabet-reduced family gives:

$$\nu = 2, r = 12 : \quad c_{12,2} = \frac{9}{34} > \frac{1}{4}, \quad K_{12,2} = 17;$$

$$\nu = 4, r = 26 : \quad c_{26,4} = \frac{19}{74} > \frac{1}{4}, \quad K_{26,4} = \frac{37}{3}.$$

As ν grows, the exponent approaches 10, compared with the AMETH floor 2 for $c = 1/4$.

11 Why the decoder does not need to authenticate inner codewords

The alphabet reduction would be much less useful if the compiler had to prove that, for every (p, j) , the vector of inner symbols was an actual encoding $\text{Enc}(u_j(p))$. That condition is nonlinear and would recreate the signed counterfeit problem. The construction avoids it entirely.

An arbitrary inner table is read only through the decoded moments (8.6). There are two cases.

- If the decoded polynomial block has generic rank at least κ , the fixed decoder yields the factorization (8.13), and actual table support pays for every surviving rank dimension.
- If the decoded block has generic rank below κ , the inner layer has disappeared from the proof. The canonical decoded moments satisfy the quotient-fold and Frobenius constraints, so the common-atom theorem and the source equations recover a Boolean source solution.

Thus inconsistent inner encodings cannot remain both cheap and semantically opaque:

$$\text{algebraically rich} \Rightarrow \text{expensive}, \quad \text{algebraically poor} \Rightarrow \text{decodable}.$$

The fixed nature of Dec is essential. If different monomials used different decoders, the common factorization (8.13) would fail.

12 Architectural optimality and routes below the constant five

The constant five has a transparent origin. Consider a compiler architecture with:

- a semantic block of $N^{\beta_D + o(1)}$ coordinates;
- honest block weight $N^{\beta_R + o(1)}$;
- $N^{\lambda + o(1)}$ sample locations;
- rank threshold κ ;
- high-rank soundness at most order κ times the sample count.

Then

$$D \asymp N^{\beta_D + \lambda} \kappa, \quad R \asymp N^{\beta_R + \lambda}, \quad S \lesssim \kappa N^\lambda,$$

so the protocol deficit satisfies

$$\frac{DR}{S} \gtrsim N^{\beta_D + \beta_R + \lambda}. \quad (12.1)$$

Mira's full ordered rank-one matrix has $\beta_D = 2$ and $\beta_R = 2$. Low-degree identifier evaluation needs one source-scale sample axis, so $\lambda \geq 1$. Therefore

$$\Delta \geq 2 + 2 + 1 = 5 \quad (12.2)$$

inside this architecture. Equation (9.9) approaches five, so the alphabet-reduced construction is asymptotically optimal under these explicit design assumptions.

This is not a universal lower bound. It identifies the deeper changes needed to improve the constant:

1. **Sparse coefficient carrier.** Replacing the full $(n + 1)^2$ outer-product block by an $N^{1+o(1)}$ carrier of honest weight $N^{1+o(1)}$ would reduce the architectural constant from five to three. Candidate mechanisms include sparse rank-one completion on the source graph, trace-dual coordinate extraction, and edge-local moment matrices with common atoms.

2. **Sample-free rank certificate.** Removing the outer sample axis while retaining an all-specializations rank-to-support theorem would reduce five to four with the full matrix carrier.
3. **Both changes.** A sparse carrier and sample-free soundness would reduce the constant to two, within a factor two of the AMETH floor.
4. **Constant one.** Reaching the AMETH constant would likely require an asymmetric witness representation in which ambient size times honest cost has only one source-size exponent beyond the high-rank gain.

13 The uniform near-square-root frontier

The conditional compiler still chooses fixed constants r and ν after a fixed exponent $c < 1/2$ is specified. It does not produce a single reduction with $c = c(D) \uparrow 1/2$.

The barrier results explain why this distinction is structural. For a polynomial-output reduction, any factor

$$\gamma(D) = D^{1/2-o(1)}$$

has protocol deficit $D^{o(1)} = N^{o(1)}$, contradicting AMETH. The stronger fixed-polylogarithmic target (1.1) has quasipolynomial protocol deficit and would collapse the exponential hierarchy even without AMETH.

Under the weaker assumption that the exponential hierarchy does not collapse, the first rank-regular dimension-only region not excluded by the present protocol is the superpolylogarithmic-loss strip

$$\gamma(D) = \frac{\sqrt{D}}{f(D)}, \quad f(D) = \log^{\omega(1)} D, \quad f(D) = D^{o(1)}.$$

A result there would exceed every fixed power D^c , $c < 1/2$, while avoiding the specific quasipolynomial-protocol collapse. It would nevertheless refute AMETH for polynomial-output reductions.

Accordingly, there are three coherent research programs:

1. improve fixed-exponent efficiency toward the AMETH constant one;
2. deliberately seek a uniform $D^{1/2-o(1)}$ theorem and accept that it refutes AMETH;
3. target $\sqrt{D}/\text{polylog } D$ while recognizing that a correct proof is simultaneously an exponential-hierarchy-collapse theorem.

14 Uniformity and verification boundary

14.1 Deterministic field construction

For fixed r and ν , all field sizes in Part II are polynomial in the source length. The small field F_0 and the outer field F/F_0 may be constructed deterministically by exhaustive irreducibility testing because ν is fixed. The degree- n identifier extension can use the same explicit binary-extension machinery as Mira's construction. The evaluation points, interpolation matrix, decoder coefficients λ_h , and every row coefficient of the binary affine system are computable in polynomial time. Power-of-two rounding and constant field slack do not alter the exponents.

14.2 What is and is not checked

The hierarchy-collapse proof and the multiplication-friendly lemma are self-contained consequences of the cited theorems and elementary finite-field algebra. The rank factorization (8.13) and exponent ledger are explicit. The following items are not currently machine checked:

- the rectangular finite-window realization step in [Theorem 7.1](#);
- the modified binary row generator and finite parameter slack;
- the final composed reduction certificate.

Mira’s original fixed-exponent theorem is formalized in Lean [\[11\]](#); the new compiler should be treated as a conditional research construction until the common-atom lemma and the revised implementation have been independently audited.

15 Conclusion

The quantitative coAM protocol exposes a universal complexity parameter:

$$\Xi = \frac{D}{\gamma^2}.$$

Driving Ξ to polylogarithmic size collapses the exponential hierarchy; driving it to $N^{o(1)}$ contradicts AMETH. For fixed-power hardness $\gamma(D) = D^c$, AMETH requires output degree $K \geq 1/(1 - 2c)$.

The multiplication-friendly alphabet reduction gives a conditional constructive counterpart. It separates large-field multiplicative semantics from small-field one-hot labels and uses one fixed decoder to preserve the high-rank factorization for arbitrary tables. Conditional on the rectangular common-atom lemma, it improves the output degree from $\Theta((1 - 2c)^{-2})$ to

$$\frac{5 + o(1)}{1 - 2c},$$

which is optimal in its dependence on c under AMETH and asymptotically optimal within the full-matrix, sampled-moment architecture.

The remaining constructive questions are now quantitatively sharp. A sparse semantic carrier can improve the constant five; a sample-free rank certificate can improve it again; and any uniform near-square-root theorem necessarily crosses a major complexity-theoretic barrier. The barrier and the construction therefore meet at the right scale: not at the unattainable endpoint of the current methods, but at the optimal fixed-exponent efficiency law.

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